CREATING A GEOGEBRA APPLET TO FIND THE UNIT DIGIT OF EXPONENTIAL NUMBERS

Aim

To create an interactive GeoGebra applet that generates random exponential questions, allows students to enter their answers, checks the answers, provides feedback, and generates new questions for repeated practice.

Procedure

Step 1: Insert the Rules for Finding the Unit Digit

- Open GeoGebra Calculator.
- Copy the image containing the rules for finding the unit digit of exponential expressions.
- Paste the image into the GeoGebra applet.

Step 2: Create Variables for the Base and Exponent

- Identify the base and exponent from the Class 7 Term 2 textbook, Example 3.11, Question 1.
- In this example, the base is 25 and the exponent is 23.
- Enter the following commands in the Input Bar:

```
x = 25
y = 23
```

Here, x represents the base and y represents the exponent. These variables will be used to display the exponential expression and generate new questions.

Step 3: Display the Exponential Question

- Select the Text tool from the toolbar.
- Open the Text dialog box by double-clicking the text object.
- Select the LaTeX option.
- Enter `\text{}` in the text field.
- Click the plus symbol and select Empty Object.
- Select the variable x to insert the base dynamically.
- Insert the variable y as the exponent.
- Click OK to display the exponential expression.
- When the values of x and y change, the displayed question will update automatically.

Step 4: Create the Answer Input Box

- Create a variable named ans and assign it the initial value zero.
- Enter the following command in the Input Bar:

  ans = 0

- Select the Input Box tool from the toolbar.
- Enter the caption as Answer.
- Link the input box to the object ans.
- Click OK to create the input box.
- Students can enter their answers directly into this box.

Step 5: Calculate the Last Digit of the Base

- Create a variable named LastDigit.
- Enter the following command in the Input Bar:

  LastDigit = Mod(x, 10)

- The Mod command calculates the remainder when the base is divided by 10.
- This gives the last digit of the base. For example, if x = 25, then LastDigit = 5.
- This variable is used to identify the repeating cycle of unit digits.

Step 6: Calculate the Cycle Length

- Create a variable named CycleLength.
- Enter the following command in the Input Bar:

  CycleLength = If(LastDigit == 0 || LastDigit == 1 || LastDigit == 5 || LastDigit == 6, 1, If(LastDigit == 4 || LastDigit == 9, 2, 4))

- This command determines the cycle length according to the last digit of the base.
- Last digits 0, 1, 5, and 6 have a cycle length of 1.
- Last digits 4 and 9 have a cycle length of 2.
- Last digits 2, 3, 7, and 8 have a cycle length of 4.
- GeoGebra automatically assigns the appropriate cycle length based on LastDigit.

Step 7: Calculate the Position in the Cycle

- Create a variable named Position.
- Enter the following command in the Input Bar:

  Position = Mod(y - 1, CycleLength) + 1

- The Mod command calculates the remainder.

- Subtracting one from the exponent and adding one to the remainder helps identify the correct position in the repeating cycle.
- This formula also works when the exponent is exactly divisible by the cycle length because it selects the last element of the cycle.

Step 8: Calculate the Correct Answer

- Create a variable named CorrectAnswer.
- Enter the following command in the Input Bar:

```
CorrectAnswer = If(LastDigit == 0, 0, If(LastDigit == 1, 1, If(LastDigit == 2, Element({2, 4, 8, 6}, Position), If(LastDigit == 3, Element({3, 9, 7, 1}, Position), If(LastDigit == 4, Element({4, 6}, Position), If(LastDigit == 5, 5, If(LastDigit == 6, 6, If(LastDigit == 7, Element({7, 9, 3, 1}, Position), If(LastDigit == 8, Element({8, 4, 2, 6}, Position), If(LastDigit == 9, Element({9, 1}, Position)))))))))
```

- This command checks the last digit of the base and selects the corresponding repeating cycle.
- The Element command selects the unit digit at the position calculated in Step 7.
- The resulting unit digit is stored in CorrectAnswer.
- This method finds the unit digit without calculating the entire exponential expression.

Step 9: Create the Check Button

- Create a Boolean variable named checked and assign it the initial value false.
- Enter the following command in the Input Bar:

```
checked = false
```

- Select the Button tool from the toolbar.
- Enter the caption as Check.
- Open the scripting section of the button.
- In the appropriate click-script box, enter the following command:

```
SetValue(checked, true)
```

- Click OK to create the button.
- When students click the Check button, the value of checked changes from false to true, allowing the applet to display feedback.

Step 10: Display Feedback for the Answer

- Create a text object to display the result of the student’ s answer.
- Enter the following conditional command:

feedback = If(checked, If(ans == CorrectAnswer, "Correct! 🎉", "Wrong! ❌"), "")

- The command checks whether the Check button has been clicked.
- If checked is true, GeoGebra compares ans with CorrectAnswer.
- If both values are equal, the applet displays Correct! 🎉.
- If the values are different, the applet displays Wrong! ❌.
- If the Check button has not been clicked, no feedback is displayed.

Step 11: Create the Next Question Button

- Select the Button tool from the toolbar.
- Enter the caption as Next Question.
- Open the scripting section of the button.
- Enter the following commands in the script box:

SetValue(x, RandomBetween(2, 99))
SetValue(y, RandomBetween(2, 99))
SetValue(ans, 0)
SetValue(checked, false)

- Click OK to create the button.
- When the student clicks Next Question, a new random base and exponent are generated.
- The answer variable is reset to zero and checked is reset to false.
- The previous feedback disappears and the new exponential question is displayed automatically.
- The variables LastDigit, CycleLength, Position, and CorrectAnswer update automatically when the base or exponent changes.

Step 12: Test the Applet

- Run the applet and observe the initial question.
- Enter the correct unit digit in the Answer input box.
- Click Check and verify that the applet displays the correct feedback.
- Enter an incorrect answer and check whether the applet displays the wrong-answer message.
- Click Next Question and verify that a new question appears.
- Repeat the process to ensure that all components work correctly.

Result

Thus, an interactive GeoGebra applet was created to display the rules for finding unit digits, generate random exponential questions, accept students' answers, calculate the correct unit digit, check the answers, display feedback, and generate new questions for repeated practice.

Rules :

Base end with	0	1	5	6
Unit digit	0	1	5	6

Unit digit of base	Base	Odd power	Even power
	4	4	6
	9	9	1

Unit digit of base	The remainder when power is divided by 4			
	1	2	3	0
2	2	4	8	6
3	3	9	7	1
7	7	9	3	1
8	8	4	2	6

Find the unit digit of 176^{168}

Answer 1

Check

Wrong! ✗

Next question